

The Correlated Anomaly Observation

Five independent measurement series, one fractional deviation band

1. The observation

Measurements made in four laboratories by three independent groups, using four different detection techniques, over intervals from a single solar flare to fifteen years, report fractional deviations that fall in the same narrow band:

$$10^{-4} \leq \frac{\delta\lambda}{\lambda_0} \leq 10^{-3}$$

The series are unrelated to one another. They measure different isotopes, at different facilities, with different instruments, for different reasons. The shared quantity is the order of magnitude of the deviation, not its sign, its phase, or its period.

2. The series

Series	Facility	Interval	Fitted α	Source
³² Si seasonal	Brookhaven	1982–1986, four years	7.9×10^{-4}	Jenkins et al. 2008
³⁶ Cl seasonal	Brookhaven	1982–1986, four years	6.2×10^{-4}	Jenkins et al. 2008
²²⁶ Ra seasonal	PTB, Germany	1983–1998, fifteen years	8.3×10^{-4}	Siegert et al. 1998
⁵⁴ Mn solar flare transient	Purdue	13 December 2006, hours	$\alpha A_{\text{flare}} = -2.5 \times 10^{-3}$	Jenkins & Fischbach 2009

³² Si/ ³⁶ Cl amplitude	Brookhaven	239 data points	$7.9(3) \times 10^{-4}$	Jenkins et al. 2008
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Fitted coupling α under the model of the time-gradient field, except where noted. The ⁵⁴Mn entry is not a bare α : the source reports the product αA_{flare} , the coupling multiplied by the flare amplitude, and that product is negative. It is listed as published and is not a like-for-like value in the column. The Brookhaven series additionally reports an annual oscillation of $7.9(3) \times 10^{-4}$ with a 35(2) day phase shift, a correlation of $r = 0.52$ against $1/R^2$, and a formal probability of random correlation $p = 6 \times 10^{-18}$. The ⁵⁴Mn transient preceded electromagnetic arrival by 36–40 hours.

3. What the values share

The four seasonal fitted couplings span 6.2×10^{-4} to 8.3×10^{-4} , a range of four across three isotopes and two facilities. The ⁵⁴Mn transient $\alpha A_{\text{flare}} = -2.5 \times 10^{-3}$ is an order of magnitude larger and is not a bare coupling. Across different isotopes, different facilities, and different physical circumstances, the couplings agree to within an order of magnitude.

No isotope-specific parameter was introduced to achieve this. The model form is fixed in advance:

$$\lambda_{\text{eff}}(t) = \lambda_0 (1 + \alpha G(t))$$

with a single coupling α per isotope and a field $G(t)$ of order 10^{-3} . The agreement across the four series is therefore a constraint on the model form, not a consequence of fitting freedom in it.

4. Reported frequencies

Power-spectrum analysis of the Brookhaven data by Sturrock et al. (2010) reports the annual frequency at greater than 5σ , a 33-day frequency at greater than 5σ , and a 12.5-year frequency at greater than 3σ . The 33-day period corresponds to the solar core rotation rate; the 12.5-year frequency corresponds to solar r-mode oscillations.

5. Companion series in other domains

The same band appears in two non-nuclear series, reported in the same corpus:

Series	Fractional deviation	Basis
Lunar Mascon residuals	10^{-4} to 10^{-3}	Residual after standard spherical-harmonic modelling
Apollo navigation residuals	10^{-4} to 10^{-3}	Velocity and guidance residuals on deep-space legs

6. Scope of this page

This page records an observation: that five independent measurement series report deviations of the same fractional magnitude. It does not claim a mechanism, and it does not present the field interpretation.

Every value above is quoted from published measurements or from the companion papers in this body of work, with the source named in each row. The interpretation of these deviations as a response to a scalar field is developed separately, in the time-gradient field model and its verification papers, and is not required to read this page.

7. Sources

[1] Jenkins, J.H. et al. "Evidence of Correlations between Nuclear Decay Rates and the Earth-Sun Distance." *Nuclear Physics A*, 807(1–2), 81–101, 2008.

[2] Siegert, H., Schrader, H., and Schötzig, U. "Half-Life Measurements of the Ra-226 Alpha Decay." *Applied Radiation and Isotopes*, 49(9–11), 1397–1401, 1998.

[3] Jenkins, J.H. and Fischbach, E. "Perturbation of Nuclear Decay Rates during the Solar Flare of 2006 December 13." *Nuclear Physics A*, 825(1–4), 101–108, 2009.

[4] Sturrock, P.A., Buncher, J.B., Fischbach, E., Gruenwald, J.T., Javorsek, D., Jenkins, J.H. "Power Spectrum Analysis of BNL Decay-Rate Data." *Astrophysics and Space Science*, 2010.

Full derivations, fitted parameters, and verification: [Time-Gradient Field Model](#) and [Mathematical Verification of the Time-Gradient Field Model](#).

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